B. Like the Bitset

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 900 | Tags: constructive algorithms, greedy, two pointers | https://codeforces.com/problemset/problem/2136/B

You are given a binary string[□] s of length n, as well as an integer k.

Aquawave wants to construct a permutation[†] p of length n, so that for each $1 \le i \le n$, where s_i=mathtt1, the following holds:

- For each interval $[l, r]$ ($1 \le l \le r \le n$) whose length is at least k (i.e. $r - l + 1 \ge k$), if it covers position i (i.e. $l \le i \le r$), then the maximum element among $p_l, p_{l+1}, \ldots, p_r$ is **not** p_i.

Note that there are **no** such constraints on indices with s_i = mathtt0.

You have to find such a permutation, or determine that such permutations do not exist.

[□]A binary string is a string where each character is either mathtt0 or mathtt1.

[†]A permutation of length n is an array consisting of n distinct integers from 1 to n in arbitrary order. For example, [2,3,1,5,4] is a permutation, but [1,2,2] is not a permutation (2 appears twice in the array), and [1,3,4] is also not a permutation (n=3 but there is 4 in the array).

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains two integers n and k ($1 \le n \le 2 \cdot 10^5$, $1 \le k \le n$) — the length of s and the integer in the statements.

The second line contains the binary string s of length n (s_i = mathtt0 or mathtt1).

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case:

- If there is at least one possible permutation:
- Print "YES" in the first line of output;
- Then, print n integers $p_1, p_2, \ldots, p_n$ ($1 \le p_i \le n$, all p_i-s are distinct) in the second line — the permutation you constructed.
- Otherwise, print "NO" in the single line of output.

You can output the tokens in any case (upper or lower). For example, the strings "yEs", "yes", "Yes", and "YES" will be recognized as positive responses.

If there are multiple answers, you may output any of them.

Example

Input

```
6
2 1
00
4 3
0010
5 2
11011
7 5
1111110
8 4
00101011
10 2
1000000010
```

Output

```
YES
1 2
YES
1 4 3 2
NO
NO
YES
6 5 2 3 4 8 1 7
YES
1 2 3 4 5 6 7 9 8 10
```

Note

In the first test case, you can output an arbitrary permutation of length $n = 2$, since all s_i-s are equal to mathtt0.

In the second test case, p=[1, 4, 3, 2] is a valid answer because:

- The only position where s_i = tt1 is $i = 3$. There are three distinct intervals [l, r] covering index 3, whose length is at least k=3: [1, 3], [1, 4], and [2, 4];
- And, for each of the three intervals, the maximum element among $p_l, \ldots, p_r$ should be $p_2 = 4$, which is not equal to $p_3 = 3$.